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Predators selectively graze reproductive structures in a clonal marine organism

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Abstract Although the fitness consequences of herbivory on terrestrial plants have been extensively studied, considerably less is known about how partial predation impacts the fitness of clonal marine organisms. The trophic role of Caribbean parrotfish on coral reefs is complex: while these fish are important herbivores, as corallivores (consumers of live coral tissue), they selectively graze specific species and colonies of reef-building corals. Though the benefits of parrotfish herbivory for reef resilience and conservation are well documented, the negative consequences of parrotfish grazing for coral reproductive fitness have not been previously determined. We examined recently grazed colonies of *Montastraea annularis* corals to determine whether grazing was positively associated with coral reproductive effort. We measured gonad number, egg number and size, and proportional reproductive allocation for grazed and intact coral colonies 2–5 days prior to their annual spawning time. We found that parrotfish selectively grazed coral polyps with high total reproductive effort (number of gonads), providing the first evidence that parrotfish selectively target specific tissue areas within a single coral colony. The removal of polyps with high reproductive effort has direct adverse affects on coral fitness, with additional indirect implications for colony growth and survival. We conclude that chronic grazing by parrotfishes has negative fitness consequences for reef-building corals, and by extension, reef ecosystems.

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Introduction

Herbivory has been intensively studied in terrestrial and marine ecosystems, and it is well known that herbivores have numerous and diverse impacts on plant fitness, community structure and ecosystem function (e.g. Lubchenco and Gaines 1981; Fritz and Simms 1992). For example, consumption of floral tissue has adverse impacts on terrestrial plant reproduction (reviewed by McCall and Irwin 2006) by directly limiting gamete production and dispersal (e.g. Louda and Potvin 1995; Krupnick and Weis 1999), as well as indirectly reducing plant fitness via reduced pollination success (e.g. Krupnick and Weis 1999; Leavitt and Robertson 2006). In clonal plants, reproductive consequences of florivory are known to include reduced seed production and increased clonal growth (e.g. Tobler et al. 2006). Furthermore, foliar herbivory (leaf damage and removal) has also been shown to negatively impact plant fitness (e.g. Wisdom et al. 1989; Avila-Sakar and Stephenson 2006). However, despite our extensive knowledge of foliar and floral herbivory in plants, comparatively little is known about the reproductive consequences following tissue removal in clonal marine invertebrates. Since colonial reef-building corals are also modular, sessile organisms that are often partially consumed (Jackson et al. 1985), coral-corallivore systems show many parallels to plant-herbivore interactions. Just as plants contribute much of the habitat structure in terrestrial environments, scleractinian corals are the major architects of tropical reefs, acting as ecosystem engineers (sensu Jones et al. 1994) to provide

the structural framework for a highly diverse assemblage of marine organisms. Direct consumption of live coral, or corallivory, represents a potentially important biotic stressor that can adversely affect coral fitness and may accelerate rates of coral decline after stress (reviewed by Cole et al. 2008; Rotjan and Lewis 2008). However, the consequences of corallivory to coral fitness have not yet been investigated.

Montastraea coral species are key reef builders in the Caribbean (Frost 1977; Budd 1988). Members of the *Montastraea annularis* species complex are hermaphroditic, and gametes develop over several months prior to the annual spawning event. Although they consume live coral as only a small component of their diet (Bruggemann et al. 1994a, b), adult parrotfish (*Sparisoma viride*, *Scarus vetula*, and *Sp. aurofrenatum*) preferentially graze on members of the *M. annularis* species complex (Garzon-Ferreira and Reyes-Nivia 2001; Rotjan and Lewis 2006), and selectively graze certain colonies over others (Rotjan et al. 2006; Rotjan and Lewis 2006). Parrotfish grazing has been shown to negatively impact coral recovery after bleaching (Rotjan et al. 2006), and to contribute to coral mortality (Gygi 1975; Rotjan and Lewis 2005). Artificial damage simulating parrotfish grazing (inflicted 10 weeks prior to spawning) has been shown to reduce reproduction in coral polyps adjacent to artificial lesions (Van Veghel and Bak 1994). However, the consequences of natural parrotfish grazing for coral reproduction have never been measured.

Like many herbivorous insects, parrotfish are known to selectively graze foods with high nitrogen and lipid contents, for example, on seagrasses (Goecker et al. 2005), reef macroborers (Rotjan and Lewis 2005), detritus (Crossman et al. 2005), and reef macroalgae (Bruggemann et al. 1994a, b). Reproductive polyps of the *M. annularis* species complex are known to have higher nutritional value than non-reproductive polyps, displaying higher dry weight and higher nitrogen biomass, which positively correlates with egg number (Szmant 1991). Reproductive gonads in *Montastraea faveolata* colonies are also known to have high lipid content (Villinski 2003). Thus, we hypothesized that parrotfish may selectively remove coral polyps with high reproductive effort.

In this study, we examined colonies of *Montastraea annularis* corals that had been recently grazed (within 2–4 weeks of sampling) by parrotfishes in order to determine whether grazing was associated with coral reproductive effort. The number of gonads, eggs, egg size, and proportional reproductive allocation were compared between intact colonies with no signs of parrotfish corallivory and colonies recently grazed by parrotfish. Within grazed colonies, we compared reproductive effort between tissue among grazed portions and intact areas of the same colony. Since partial predation has important reproductive consequences in terrestrial plants, we conducted the present study to understand the reproductive consequences of chronic parrotfish grazing on tropical corals, and how corallivory might resemble terrestrial herbivory.

Methods

Reproductive ecology of *Montastraea annularis*

In the *Montastraea annularis* species complex, oogenesis begins at least 4 months prior to spawning and spermatogenesis requires at least 10 weeks (Szmant 1986; Van Veghel 1994). Both direct field and histological observations confirm that spawning in the *M. annularis* species complex usually occurs in late August or early September, approximately 1 week after the full moon (Szmant 1991; Knowlton et al. 1997; Sanchez et al. 1999). Spawning is locally synchronous, and colonies within a region usually release gamete bundles within minutes of each other (Knowlton et al. 1997; Sanchez et al. 1999; Levitan et al. 2004). These broadcast spawners release gamete bundles containing both eggs and sperm that rise through the water column and separate at the surface, where fertilization occurs (reviewed by Richmond 1997).

Sampling design and collection

This study was conducted at 6–9 m depth in the shallow upper spur and groove habitat (described in detail by Rüetzler and Macintyre 1982) during late August 2005 at Southwater Caye (16°46.575N, 88°04.494W) on the Belizean barrier reef. *Montastraea annularis* corals usually spawn 7–8 days after the full moon in August (Knowlton et al. 1997). To ensure ripe gametes, we collected small coral samples (described below) 2–5 days prior to the major spawn, from August 21 to 24. Since egg production may be affected by disease or through direct contact with another coral, we chose isolated, healthy colonies. Before collection, the length and width of each *M. annularis* colony was measured to the nearest 1 cm using a flexible ruler. *M. annularis* corals with a surface area less than 50 cm^2 are generally not reproductive (Szmant-Froelich 1985; Van Veghel and Kahmann 1994); therefore, we chose colonies ranging in size from 176 to 9,020 cm^2.

In order to compare reproductive effort from intact colonies versus those recently grazed by corallivorous parrotfishes, we haphazardly sampled clusters containing pairs of similarly sized intact and grazed *M. annularis* colonies located within 1 m of each other (Fig. 1). Colonies were considered grazed if they had more than six pairs (showing upper and lower jaw marks) of recent (within ~2–4 weeks) parrotfish grazing scars. Recent scars

appeared white due to skeletal exposure if no visible regrowth, or due to lack of zooxanthellae where slight tissue regeneration may have occurred (Fig. 2a). On most scars, no tissue was observed, with little to no algal turf colonization or biofilm formation, suggesting very recent grazing (within days). On grazed colonies, the number of bites ranged from 11 to 135 ($\bar{X}$bite density = 4.86 bites/dm^2 coral tissue). Grazing scars were likely made by *Sparisoma viride* parrotfish species (Rotjan and Lewis 2006).

From each grazed colony, we first sampled intact tissue from among the fresh grazing scars (G_g = grazed portion of the grazed colony), and then sampled intact tissue located approximately 15 cm away from the grazed area (G_i = intact portion of the grazed colony). We also sampled tissue from a nearby (~1 m away), similarly sized, intact colony (I = intact colony tissue); samples were taken from equivalent locations on each colony and were always taken from central locations (never within ~15 cm of the growing edge). This design was chosen so that we could examine both between-colony differences (G_i vs. I), and within-colony differences (G_g vs. G_i). We collected samples from 24 clusters for a total of 72 samples; each cluster contained G_i and G_g samples from the same colony, and an I sample that was paired with the previous samples by location and size (Fig. 1).

Small samples (~5 cm^2) from each colony were collected with a hammer and chisel, and fixed in 4:1 seawater-Z-Fix® solution (Anatech Ltd., Battle Creek, MI) in the field. Coral samples were moved to fresh solution after 24 h and wrapped in fixative-soaked paper towels for transport. Z-fix is a zinc-based fixative recommended for preservation of scleractinian coral tissue (E. Peters, personal communication). In the lab, corals were rinsed briefly in de-ionized (DI) water and placed in fresh fixative solution for at least 2 weeks. Coral samples were decalcified using 2.5% HCl, 0.1% EDTA solution that was changed daily, and the resulting soft tissue was preserved in fixative until dissection.

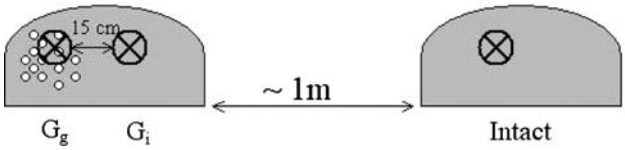

Fig. 1 Sampling schematic cartoon showing a single cluster of *Montastraea annularis* corals sampled 2–5 days prior to spawning in August 2005. *Crossed circles* represent ~5 cm^2 tissue samples collected with a hammer and chisel. *Smaller open circles* represent recent parrotfish grazing scars (grazed within 1 week of sampling). Fifteen clusters (blocks) were sampled, each consisting of a grazed *M. annularis* colony taken among recent grazing scars (G_g), a sample of intact tissue taken 15 cm away from grazing scars from the same colony (G_i), and a sample taken from a similarly sized, nearby colony with no grazing scars (intact)

Coral dissections

We examined reproductive output of 10 randomly chosen polyps from each of 72 samples; polyps were individually numbered and then selected using a random-number table. Each selected polyp was dissected out of the colony and stained with a 1% toluene blue solution in DI water for 3–5 s and then rinsed, which helped to distinguish eggs and gonads from remaining tissue. Polyps were dissected under a stereoscopic microscope and polyp diameter and height were measured to estimate polyp volume (Fig. 2b). For all 720 polyps, we counted the total number of gonads (Fig. 2c) and the number of eggs within each gonad (Fig. 2d). Identification of reproductive structures followed Szmant (1986) and Villinski (2003), and were confirmed by E. Peters and A. Szmant (personal communication). For a subset of the 72 samples collected (subset $N = 39$ samples totalling 234 polyps representing 13 clusters with eggs visibly present in all samples (G_g, G_i, and I)), we also carefully isolated and photographed every other gonad to estimate egg size. Projected egg area (mm^2) was determined from digital images using Matlab 6 (Mathworks Co.) for five randomly chosen eggs per fecund polyp and up to 10 polyps per sample, and was used to estimate female reproductive quality (defined below). To estimate egg volume (mm^3) for total reproductive allocation measurements, we multiplied egg area by a standardized egg height (0.015 mm) estimated from the mean height of 10 eggs within a single colony and modeled each egg as a cylinder. Although *M. annularis* are hermaphroditic (Szmant 1986), we did not attempt to quantify spermaries because they are small and translucent, and therefore difficult to identify without histological staining and sectioning (E. Peters, personal communication).

Total reproductive effort for each sample was measured as the total number of gonads summed across 10 randomly selected polyps (gonads contain both eggs and spermaries). Total reproductive effort represents the largest energetic investment, since gonads contain both eggs and spermaries, in addition to gonad membranes and other potentially nutritive tissues within. **Female reproductive effort** for each sample was based on the total number of eggs summed across 10 polyps. Two additional aspects of female reproductive output were examined: (1) egg size (area) as a measure of female reproductive quality, and (2) number of eggs per gonad, calculated by averaging the number of eggs with each gonad for each polyp, again averaged over 10 polyps for each sample. We also estimated **total reproductive allocation** for a subset of 13 clusters by calculating total egg volume as a percentage of

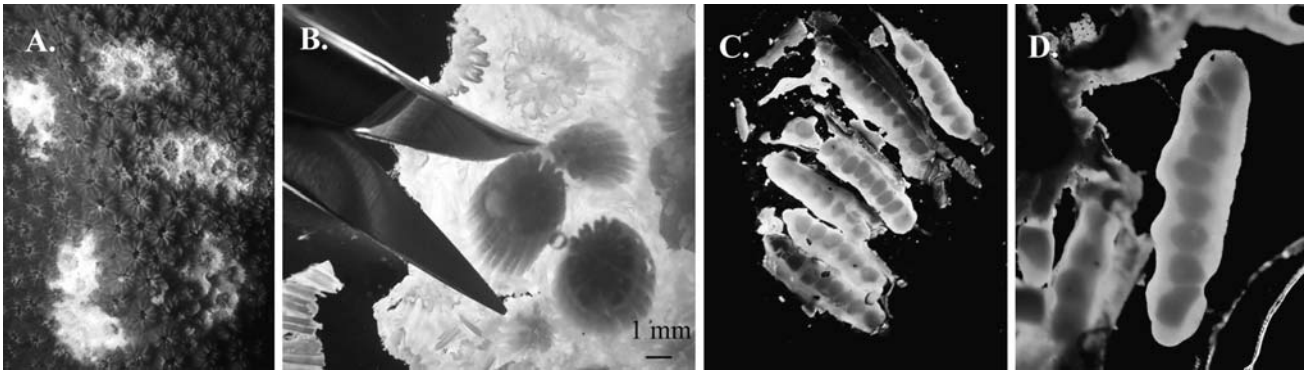

Fig. 2 *Montastraea annularis* corals sampled 2–5 days prior to spawning. Photographs illustrate grazing scars on a colony in the field (**a**, photo by A. Bruckner), and the colony underside showing individual polyps after decalcification (**b**), six representative gonads dissected out of a single polyp (**c**), and close-up of multiple eggs inside each gonad (**d**). Average egg area is 0.054 mm^2. Gonads also include spermaries (not visible)

the polyp volume (ratio of these two volumes multiplied by 100).

In order to examine differences in number of gonads, eggs, and eggs per gonad between polyps from intact (I) and grazed (G_g and G_i) coral samples, we used one-way repeated measures ANOVA (SYSTAT 11) with post-hoc multiple comparison tests for repeated measures. This allowed us to compare reproductive effort between intact coral tissue from two *M. annularis* colonies, where we compared grazed (G_i) versus intact (I) tissue, as well as reproductive effort within grazed colonies (G_g vs. G_i). We used a repeated measures approach using location (cluster) as a blocking factor, since many location-related variables (depth, light intensity, water flow, etc.) may potentially affect coral reproduction. In other words, instead of repeated observations made on individual units through time, we have followed another conventional repeated measures, or split-plot, approach, using a groups by trials design where we used a treatment structure (in our case, clusters) to maintain the relationships between coral samples in space (Quinn and Keough 2002). All data met the assumptions of each analysis. We similarly used a repeated measures approach to examine mean egg size and total reproductive allocation (total egg volume as a percentage of polyp volume) per colony. Only a subset of colonies (13 clusters) was examined; clusters with no eggs in any given category (I, G_i, or G_g) were eliminated from egg size analysis.

Results

We found significant differences in total reproductive effort (total number of gonads, which contain both spermaries and eggs, per 10 polyps) among *M. annularis* samples taken from the intact portion of grazed colonies, the grazed portion of grazed colonies, and intact colonies (Fig. 3a, b, one-way repeated measures ANOVA, $F = 3.29$, $df = 2, 46$, $p = 0.0467$). Specifically, we found fewer gonads in intact portions of grazed colonies (G_i) compared to grazed portions of the same colony (G_g) (repeated measures post-hoc multiple comparison $d = 16.61$, $p = 0.0345$). These intact portions of grazed colonies (G_i) also had fewer gonads compared to intact colonies (repeated measures post-hoc multiple comparison $d = -24.13$, $p = 0.0214$). Thus, within grazed *M. annularis* colonies, parrotfish grazing was associated with areas showing higher total reproductive effort, even though parrotfish grazing was not observed on nearby colonies with similar high reproductive effort. Number of gonads did not differ between intact (I) and grazed (G_g) samples (repeated measures post hoc multiple comparison $d = -7.52$, $p = 0.5146$).

Female reproductive effort (total number of eggs per sample) did not significantly differ across categories (Fig. 3c, d, I vs. G_i vs. G_g; one-way repeated measures ANOVA $F = 2.10$, $df = 2, 46$, $p = 0.1342$). Similarly, total reproductive allocation of *M. annularis* polyps (egg volume as a percentage of total polyp volume) was not significantly different across categories (Fig. 4; one-way repeated measures ANOVA $F = 1.0387$, $df = 2, 24$, $p = 0.3693$). The mean number of eggs per gonad did not vary across categories (one-way repeated measures ANOVA $F = 0.29$, $df = 2, 46$, $p = 0.7456$ (Fig. 3e, f). Although there was considerable variation in egg area (range = 0.006–0.262 mm^2), egg size (area) did not vary among categories ($\bar{X}mm^2 \pm$ SE for $G_g = 0.049 \pm 0.006$; $G_i = 0.059 \pm 0.006$; $I = 0.055 \pm 0.005$; one-way repeated measures ANOVA $F = 1.3965$, $df = 2, 24$, $p = 0.2668$).

Discussion

Even in plants, where the impacts of herbivory on plant biomass and fitness have been comprehensively studied for

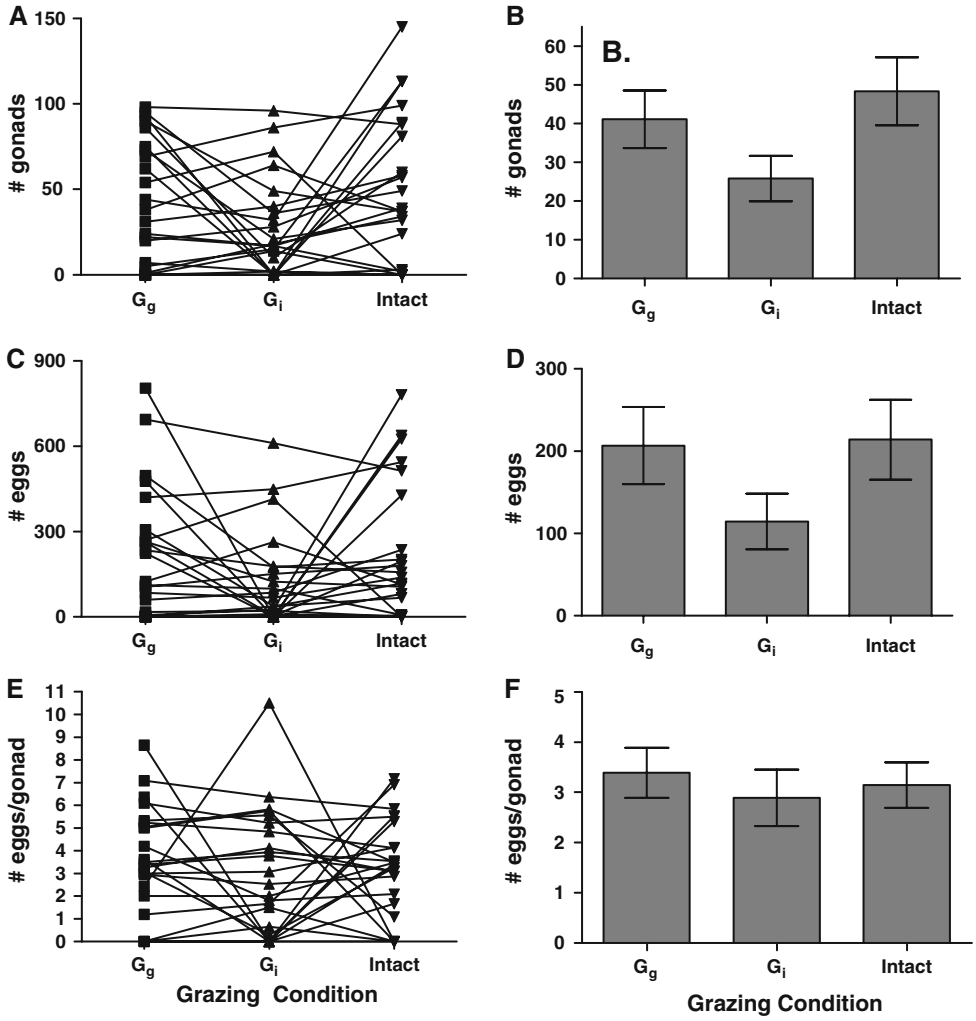

Fig. 3 Reproductive variables measured on *Montastraea annularis* corals collected from tissue adjacent to recent parrotfish grazing scars (G_g), intact portions of the same grazed colony (G_i), or tissue from a nearby, non-grazed colony (intact). Reproductive effort is represented by the total number of gonads summed across 10 polyps (**a**, **b**), the total number of eggs per 10 polyps (**c**, **d**), and mean number of eggs per gonad (**e**, **f**). For each variable, *left panel* shows data by cluster, with G_g and G_i from the same colony (paired by genotype), and *I* representing a similarly sized, nearby colony (paired by location). *Right panel* shows overall mean ± 1 SE (not grouped by cluster). $N = 24$ clusters; 3 colonies per cluster; 720 polyps examined

decades (e.g. Marquis 1984; Cyr and Pace 1993), the impacts of herbivores on leaves, flowers, and roots continue to be widely investigated. In marine invertebrates, there are comparatively few studies examining the fitness impacts of partial predation, despite the clear parallels between herbivory and corallivory. The present study makes several new contributions to our understanding of reproductive tissue removal in clonal marine invertebrates. First, we show that polyp corallivory can directly reduce coral reproductive fitness by consuming gamete-laden polyps, which contain both eggs and spermaries. Within *Montastraea* spp. coral colonies, we demonstrate that parrotfish selectively graze on polyps with high reproductive effort (number of gonads), with the same trend for egg number. This is the first demonstration that parrotfish corallivory is non-random and appears to be directed at coral tissue areas with the largest reproductive investment (number of gonads). Gonads contain eggs, but also contain added value in the form of spermaries, gonad membranes, and other potentially nutritive tissues, thus making gonad number an important measure of reproductive fitness and nutrition. Our results also suggest that chronic parrotfish corallivory may have indirect reproductive fitness consequences for corals, since polyps located ~15 cm away from grazing scars showed lower reproductive effort and reproductive allocation.

Within grazed colonies, we found significantly higher total reproductive effort (number of gonads) in coral tissue sampled from among grazed areas (G_g) compared to intact tissue from the same colony (G_i) (Fig. 3a, b). Female reproductive effort (Fig. 3c, d) followed the same trend, although not significant, likely due to the variance in number of eggs per gonad (Fig. 3e, f). The most likely explanation for higher total reproductive output (gonads) in polyps located among grazing scars is that parrotfish selectively target coral tissue areas that show high reproductive effort and allocation. An alternate explanation might be reproductive compensation, whereby a plant shows increased reproductive effort in response to herbivory, which has been observed in terrestrial plants (reviewed by Harris 1974; shown by Dyer 1975; Paige and Whitham 1987; Peterken and Conacher 1997; Paige et al.

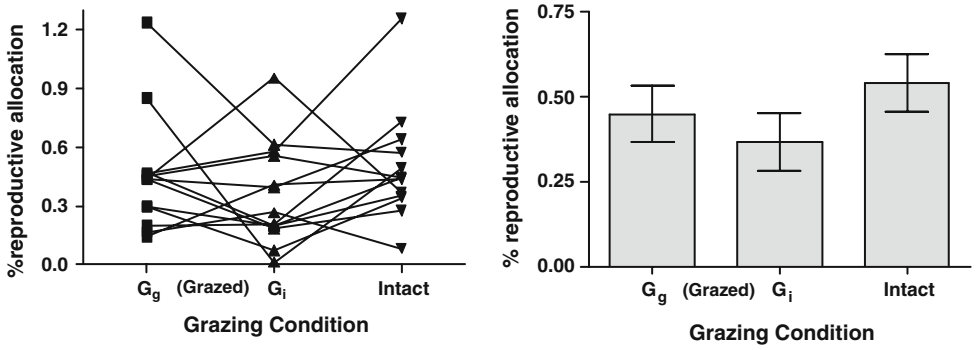

Fig. 4 Average total reproductive allocation (egg volume as a percentage of total polyp volume) of coral polyps from grazed *Montastraea annularis* colonies sampled near areas recently grazed by parrotfishes (G_g); polyps sampled from intact tissue on grazed colonies (G_i), and polyps sampled from nearby intact colonies (intact). Reproductive allocation was calculated by total egg volume as a percentage of polyp volume. *Bars* represent means ± 1SE ($n = 39$ colonies (13 clusters)). *Left panel* shows data by cluster, with G_g and G_i from the same colony (paired by genotype), and *I* representing a similarly sized, nearby colony (paired by location). *Right panel* shows overall mean ± 1 SE. (not grouped by cluster)

2001). However, this explanation is unlikely to apply to corals because they require at least 4 months to complete oogenesis (Szmant 1986; Van Veghel 1994), and sampled colonies were only recently grazed (within 2 weeks); thus, there would have been insufficient time for reproductive compensation to occur. Parrotfish may selectively target tissue areas with high reproductive effort and allocation because of their high nutritional value; gonads are both lipid- and nitrogen-rich (Szmant 1991; Villinski 2003). *Sparisoma radians* parrotfish have been similarly shown to distinguish nitrogen-enriched *Thalassia testudinum* seagrass blades from control blades (Goecker et al. 2005). Although the mechanism for detection remains unknown, parrotfish have also been shown to preferentially consume high-nitrogen macroborers (Rotjan and Lewis 2005) and relatively nitrogen-rich macroalgae (Bruggemann et al. 1994a, b).

Our results suggest that parrotfish are responding to existing within-colony variability in reproductive effort and allocation. Such spatial variability in reproductive effort has been previously described in *Montastraea* spp. colonies; border polyps (at the growing edge of a colony) have lower fertility and fecundity than centrally located polyps (Szmant 1991; Van Veghel and Kahmann 1994). Ours is the first study to show spatial variation in reproductive effort among centrally located polyps. Grazing incidence and extent are also patchy on different spatial scales; both at the between-colony and between-site levels (Rotjan and Lewis 2006). Spatial variation in reproductive output is also found in plants; for example in *Catalpa speciosa* trees, fruit production is localized to certain branches and fruits are spatially patchy within a tree due to patterns of herbivory and defoliation (Stephenson 1980). The factors generating such spatial variation in corals remain unknown, although reproductive patchiness could potentially be a consequence of prior or chronic parrotfish grazing from months or even years past. It is known that that parrotfish cause chronic damage by repeatedly grazing particular coral colonies (Bruckner and Bruckner 1998; Bruckner et al. 2000; Sanchez et al. 2004; Rotjan and Lewis 2005; Rotjan 2007). Over the course of several months, such damage may result in lower reproduction by polyps surrounding grazed areas. For example, artificial mechanical damage to corals has been shown to reduce reproduction in neighboring polyps for months afterwards, because neighboring polyps divert energetic resources to tissue regeneration instead of reproduction (Kojis and Quinn 1981; Kojis and Quinn 1985; Rinkevich and Loya 1989; reviewed by Harrison and Wallace 1990; Van Veghel and Bak 1994). Other possible explanations for reproductive patchiness might include patchy colony susceptibility to disease, bleaching, or other stressors, but these possibilities have yet to be investigated. Thus, the causes for within-colony spatial variability in reproductive effort remain unknown, but are likely to reflect prior damage that is localized to particular colony areas.

Selective targeting by parrotfish of scleractinian coral polyps with a high density of mature gonads is likely to have major consequences for corals, and our findings are consistent with the only other study examining reproductive structure density as a cause for corallivory in which *Chaetodon capistratus* butterflyfish were found to selectively graze reproductively mature *Plexaura* spp. gorgonian coral colonies (Lasker 1985). It has also been anecdotally noted that Caribbean parrotfish grazing incidence and intensity on *Montastraea* spp. scleractinian colonies increases just prior to spawning (A. Szmant, communicated in Miller and Hay 1998; Rotjan 2007), though parrotfish are known to graze live coral year-round (Sanchez et al. 2004; Rotjan 2007). In addition to directly

removing the most fecund polyps, our results therefore suggest an indirect effect whereby year-round parrotfish grazing, which has been previously shown for artificial mechanical damage and fragmentation, may reduce the future reproductive effort of neighboring polyps (Kojis and Quinn 1981; Kojis and Quinn 1985; Szmant-Froelich 1985; Rinkevich and Loya 1989; Harrison and Wallace 1990; Van Veghel and Bak 1994). The cause of selective corallivory in non-reproductive seasons remains unknown.

Previous studies have shown that artificial damage affects coral reproductive potential (reviewed by Henry and Hart 2005). Van Veghel and Bak (1994) conducted an elegant investigation of *Montastraea annularis*, *M. faveolata*, and *M. franksi* corals and found that colonies inflicted with artificial lesions ~10 weeks prior to spawning showed reduced reproductive effort (fewer gonads per polyp and fewer eggs per gonad) in neighboring polyps compared to polyps located farther from the lesion area on the same colony. Their design allowed a 10-week period for coral polyps to respond to the experimentally induced, artificial damage; our study design eliminated the possibility of polyp response to naturally occurring corallivory prior to spawning. Furthermore, they found that newly regenerated polyps had no reproductive activity, and polyps near regenerating tissue often had eggs of differing sizes (mature and immature), whereas distant polyps had only mature eggs (Van Veghel and Bak 1994). In terrestrial plants, simulated herbivory does not mimic effects of natural herbivores, as herbivore chemical or behavioral cues are lacking. Differing impacts of natural versus simulated corallivory have not yet been evaluated, but our study is the first to demonstrate reproductive consequences by naturally occurring corallivory. Together with previous studies artificially simulating damage, our results suggest that corallivory not only directly reduces coral reproduction by removing polyps with high levels of mature eggs and gonads, but also indirectly reduces reproduction in neighboring polyps that must devote energy to tissue regeneration rather than reproduction. Thus, parrotfish grazing has important implications for colony fitness beyond tissue removal alone.

We also found high levels of reproductive effort in intact colonies that showed no signs of parrotfish grazing. These colonies should also have high nutritional content, based on their high density of lipid-rich eggs and gonads (Szmant 1991; Villinski 2003). One possible explanation for parrotfish avoidance is that these coral colonies, like many terrestrial and marine plants, may have effective chemical or structural defenses against grazing. However, although many studies have suggested the possibility of chemical defenses in scleractinian corals (reviewed by Harvell 1999), so far there is no evidence for chemical or mechanical defensive mechanisms as successful deterrents of parrotfish predation (Rotjan 2007). Thus, explanations for observed *Montastraea annularis* colony-to-colony variation in parrotfish grazing susceptibility remain to be identified.

Coral reefs are among the most ecologically diverse and economically important ecosystems in the tropics, and are suffering rapid declines due to both natural and anthropogenic stressors (reviewed by Gardner et al. 2003; Lesser 2004). Much recent work has been devoted to understanding factors that influence coral larval recruitment and settlement success (e.g. Shearer and Coffroth 2006; Wilson et al. 2006; e.g. Kitamura et al. 2007), however, few studies have examined how stressors are impacting the overall reproductive output of corals. This study contributes to our understanding of the fitness consequences, beyond tissue removal, of parrotfish corallivory. As corals continue to decline (Gardner et al. 2003) and corallivorous parrotfish populations locally increase (Rotjan and Lewis 2006, 2008), natural trophic interactions may become imbalanced and may shift to new roles in coral reef ecosystems. Just as terrestrial herbivory can influence the reproductive performance of habitat-forming plants, marine corallivory can negatively impact the reproductive fitness of reef-building corals. Integrating clonal marine invertebrates into the context of plant-herbivore interactions is essential because, like herbivores, colony grazers have the potential to influence the evolution of defensive traits and colony characteristics.

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